

Report Guidelines

BCHM 4354 - Biochemical Communication - Spring 2025

Overview of the project report

You will write a report (7 pages max) detailing the hypotheses, experimental approaches & computational methods used to answer the following scientific question:

What is the chemical basis of mosquito attraction to different hosts?

Expectations (in no order of importance)

- Cite the relevant literature, and include a reference section at the end of your write-up (not included in the 7-page limit).
- State your hypotheses, alternate hypotheses, methods and summarize the experiments.
- Type your report:
 - Font: Times New Roman, Arial (or similar).
 - Size: 12 pts, single-spaced.
 - Numbered pages.
 - 7 pages **MAX**. Anything beyond the end of page #7, will not be considered.
 - Figures should be positioned within the text and count toward the 7 page limit.
 - Any word processor is accepted, as you will submit your report in PDF format.
 - If written in **R markdown**, submit the rendered PDF **and** your “.Rmd” file. (FYI, these guidelines have been written in **R markdown**)
- You need to embed figures in the document (in addition to displaying the results of your analysis, a figure can also be helpful for describing a particular technique, method, concept, or experimental design).
- Plagiarism will be checked.
- Equal contribution of **all** authors is expected. It is your responsibility to develop a strategy allowing everybody to use *R*. Each group is asked to complete a collaboration plan and submit it to the instructor. Be creative, teach each other, and report your strategy as well as the contribution of each student in the *Author contribution* section of the report. Attendance will be checked by the instructor and if you need to miss a session for any particular reason, it is your responsibility to catch up with your group and make up for the missed time. Attendance counts toward the final grade for the report.

Outline

Introduction

Present the broader context of the question and describe the current state of knowledge in the field. State the question(s) and present your hypothesis(-es).

Material and methods

Briefly describe the experimental methods used to collect the data (~2-3 sentences). Then, describe the methods you used to arrange, transform, analyze and visualize the data. Name and cite the key *packages* and *functions* that allowed you to reach your conclusions. Present *and justify* the statistical tests you used. Figures and schematics can be used to help explain your methods.

Results

Present the results of your analysis. Make and embed figures depicting the results of your analysis. In the text, include the details and outcome of your statistical analysis (e.g. name of tests, sample sizes, *p*-values), and make sure to make references to your figures. For example, this is a reference to my first figure: (**Figure 1**). Also, in Rmarkdown, you can hide code chunks by adding “echo=FALSE” to a code chunk. See the Rmd file for this document as an example.

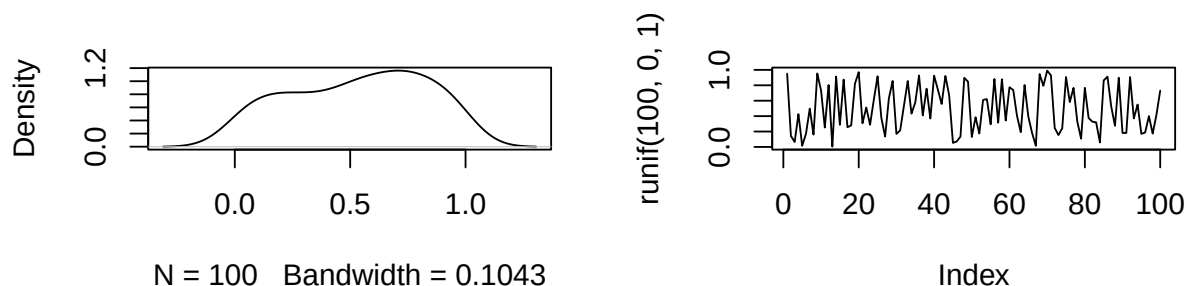


Figure 1: Plotting and figure example. Don't forget to include figure captions!

Discussion

In this last section of the report, interpret, synthesize and summarize your results. How do they inform or answer your question? Interpret your results and discuss their implication in a broader context. Cite relevant literature and discuss potential limitations of your study. Finally, conclude with a discussion of future directions for the field, or new research avenues opened by your work.

Authors contribution

Not included in the 7 pages. Describe the contribution of each member of your group and explain the strategy you employed to teach each other and make sure you all learned how to use R. If a (some) member(s) of your group missed in-class sessions, explain how you scheduled time off-class to make up for it.

References

Not included in the 7 pages. Use articles from the lectures and discussion readings as examples of reference formatting. In the main text, make reference to the literature you cite. Here, list the complete references of these articles. Make sure the format is consistent throughout your report and your reference list.

Script

Not included in the 7 pages. Provide the R script used to generate the figures and statistical analyses presented in your report. Using # add comments describing your approach. When running your final script, the instructor needs to be able to generate the *exact* figures of your report.

Dataset

Scientists collected the volatile organic compounds (VOCs) emitted by 4 human volunteers washed with different brands of soap. VOC collection was achieved by having volunteers wear nylon sleeves on their forearms for 1 hour. The sleeves were then used for “dynamic head space sampling” of the VOCs, followed by Gas Chromatography - Mass Spectrometry (GCMS) analysis. After correction for the sampling duration, and removal of contaminants identified from their control treatments, the team identified a total of 85 VOCs produced by the soap-washed hosts. Each chemical’s abundance (in *ng*) is provided in the data set, with 4 replicates for unwashed volunteers and 3 replicates for each volunteer-soap combination.

For each combination of soap and volunteer, researchers also used a separate set of nylon sleeves to perform behavioral assays on *Aedes aegypti* female mosquitoes to determine their behavioral response to the scent. In these assays, groups of mosquitoes were released into a cage containing two cups: one loaded with a sleeve worn by a volunteer, and one loaded with a clean, control sleeve. Mosquito attraction was quantified as a preference index comprised between +1 (*i.e.*, high level of attraction; all the tested mosquitoes are attracted) and -1 (*i.e.*, high level of repulsion; all mosquitoes preferred to land on a control, unscented sleeve). This information is provided as the last row of the data set.

Months after the study was completed, the team sampled a fifth volunteer and analyzed their body odor composition, but did not have the resources to conduct a behavioral assay with mosquitoes. This new sample is provided as the last column of the data set.

Using your newly acquired ability to manipulate and analyze data sets, the team is counting on you to determine the following:

Is mosquito attraction influenced by the scent composition of the host?

Is there an influence of the type of soap used?

Using predictive modeling, can you predict how mosquitoes will react to the scent of the fifth volunteer?

A multi-step analysis

In order to answer these questions, you will have to go through multiple steps:

1. Manipulate the data set so that it can be used in your analysis (consider **long vs wide** formats, sub-setting, etc).
2. Provide descriptive statistics to characterize odor profiles.
3. Use inferential statistics to compare and contrast odor profiles.
4. Perform a multivariate analysis (*e.g.*, Principal Component Analysis or NMDS) in order to visualize similarities between the chemical composition of each soap-volunteer combination.
5. Do you see a pattern? Are some samples clustering together?
6. Use machine learning (*e.g.*, Support Vector Machine) to predict mosquitoes’ preference for the new volunteer.

A group project!

This assignment is a **group** project. Work together to organize your time, distribute tasks, keep track of progress, help and teach each other, search for literature, and share the workload! You have until the end of the semester to produce a detailed 7-page report of your work. Clearly define your hypotheses. Present your methods, results and interpretations. Collaboration between groups and class-wide discussions are encouraged!

Raw data

The raw data set is available on Canvas. Download the file, save it in the same folder as your R script and load the data by including the following lines of code in your script:

```
library(tidyverse)
library(readxl)

setwd(dirname(rstudioapi::getActiveDocumentContext())$path))
Raw.data <- read_excel("Dataset_R_2025.xlsx")
```

R markdown

These guidelines were designed in R **markdown** and the source code is available on Canvas.